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Serial dependence is stronger for peripheral than for central vision

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Abstract
Serial dependence in vision refers to the fact that perceptual judgements are biased by earlier experiences, and has been thought to reduce sensory uncertainty and sustain perceptual continuity over time and space. While vision changes with eccentricity, little is known about if and how serial dependence differs in the periphery relative to fovea. Here we aimed to reduce this gap by comparing serial dependence for centrally and peripherally presented stimuli. Experiment 1 presents a reanalysis of an existing dataset from an earlier working memory task requiring the memorization of differently oriented gratings, presented either centrally or at 15° eccentricity. Experiment 2 also varied pre-knowledge of the item's location through spatial cueing. Experiment 3 replicated Experiment 1 but with lower contrast levels and equating the probabilities of central and peripheral stimuli. Across all experiments we observed an attractive bias towards the orientation of the preceding trial at all locations. Crucially, this bias was always larger in the periphery relative to the central position, and it was mainly the current item's location that drove this effect, rather than the previous item's location. Pre-knowledge of item location failed to influence the eccentricity effect serial dependence, nor did reduced contrast or differential probabilities change the conclusions. Our results thus demonstrate that serial dependence is not equal across eccentricity. The data and the scripts are available at: https://osf.io/v56hn/?view_only=6d4d5bba493b4bc788c3eed8decd8370

Keywords Serial dependence · Foveal vs. peripheral vision · Perception · Visual working memory

Introduction

Perceptual decisions do not solely depend on sensory input but are also shaped by prior percepts—a phenomenon known as *serial dependence* (Cicchini et al., 2014; Fischer & Whitney, 2014). Serial dependence is typically characterized by an attraction of current perception towards previously encountered stimuli, although repulsive biases have also been documented (Fritsche & de Lange, 2019; Fritsche et al., 2017). While the exact mechanisms are subject to debate, it is generally agreed that serial dependence may occur at multiple levels of the visual stream and serve to improve perceptual decisions by reducing uncertainty and increasing the continuity or stability of sensory representations in variable, noisy input (Braun et al., 2018; Ceylan et al., 2021; Cicchini et al., 2018, 2021; Collins, 2019; Fischer & Whitney, 2014; Fritsche et al., 2020; John-Saaltink et al., 2016; Manassi & Whitney, 2024; Manassi et al., 2023; Pascucci & Plomp, 2021; Pascucci et al., 2023; Samaha et al., 2019; Suárez-Pinilla et al., 2018; van Bergen & Jehee, 2019).

Importantly, however, the visual field is not homogenous across eccentricity, and differences between foveal and peripheral vision emerge as early as in the retina, with distinct distributions of photoreceptor types (Curcio et al., 1990). Due to stronger convergence at the level of ganglion cells with increasing eccentricity, sensory information is pooled over larger parts of the visual field, resulting in larger receptive field sizes, and reduced areas of visual cortex relative to areas dedicated to foveal input (Curcio & Allen, 1990; Dumoulin & Wandell, 2008; Horton & Hoyt, 1991; Rovamo & Virsu, 1979). Consequently, visual acuity is impoverished in the periphery, susceptibility to interference by visual crowding increases, and perception becomes

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more tuned to the statistical characteristics of the input than to individual features (e.g., Freeman & Simoncelli, 2011; Levi, 2008; Pelli, 2008; Pelli & Tillman, 2008; Rosenholtz, 2016; Strasburger et al., 2011; Whitney & Levi, 2011). In addition to these feedforward effects, feedback-dependent processes also appear disadvantaged for peripheral vision, although less is known about this. Attention favours more central stimuli (van Heusden et al., 2021, 2023, 2024; Wolfe et al., 1998), and increased eccentricity comes with reduced attentional resolution (Intriligator & Cavanagh, 2001), resulting in lower discrimination performance even when differences in sensory input are controlled for by cortical magnification of peripheral stimuli (Carrasco et al., 2003; Staugaard et al., 2016; see Olivers, 2025, for a comprehensive review on attention and eccentricity).

Both from the perspective of providing perceptual continuity and from the perspective of reducing perceptual or decisional uncertainty we may expect serial dependence to differ for central and peripheral vision. On the one hand, the stronger pooling and therefore overlap of feature signals in the periphery is expected to make for stronger continuity between consecutive perceptual instances. On the other hand, the reduced precision and increased uncertainty that comes with peripheral vision would increase the reliance on preceding interpretations to arrive at stable inferences. Note that these perspectives are not mutually exclusive as both may occur and in fact one may provide a solution for the other. Yet to our knowledge no studies have systematically assessed the effects of eccentricity on serial dependence. Several studies have employed stimuli at different eccentricities, but did not analyse for eccentricity effects or simply assumed equivalence in terms of attractive biases (Collins, 2019, 2022; Manassi et al., 2023; Fischer & Whitney, 2014). Collins (2019) attempted to control for eccentricity effects by applying the cortical magnification factor to her stimuli. Rather than trying to compensate for eccentricity effects, in the current study we were explicitly interested in how serial dependence changes for peripheral versus central stimuli.

To this end, we assessed serial dependence effects for stimuli at the fovea and in the close periphery (15 dva), and contrasted the effect of eccentricity on both inducing the bias as well as being affected by it. Figure 1 illustrates the main procedure. On each trial, a single orientation pattern was presented at one of three locations separated by 15 dva along the horizontal axis: left peripheral, central, and right peripheral. Observers remembered the orientation for a delayed continuous recall test at the end of the trial, where they indicated it using a joystick. Here we focused the analysis on the behavioural responses as a function of the orientation on the preceding trial. Experiment 1 consisted of a reanalysis of a dataset that we originally collected for an earlier study using this procedure (and which focused on a different question; Kandemir & Olivers, 2024). To replicate and extend the main findings, Experiment 2 then adopted a similar design, while also seeking to manipulate attention, using a cue prior to encoding. Given that attention has been shown to modulate serial dependence effects across space, and given that attention may not be distributed evenly across eccentricity (Olivers, 2025), we sought to explore if, and how precueing attention might modulate the difference in serial dependence between central and peripheral vision. In Experiments 1 and 2, peripheral stimuli (combining left and right locations) were more prevalent than central stimuli, which may have affected the pattern of findings. Experiment 3 controlled for this, and moreover used lower contrast stimuli typical for serial dependence research. To foreshadow the main finding, in all experiments we observed stronger serial dependence for more peripheral stimuli.

Experiment 1

Methods

Participants

The participant sample consisted of 30 healthy young adults (23 women, $M_{\text{Age}} = 20.0$ years, $SD_{\text{Age}} = 1.81$) who

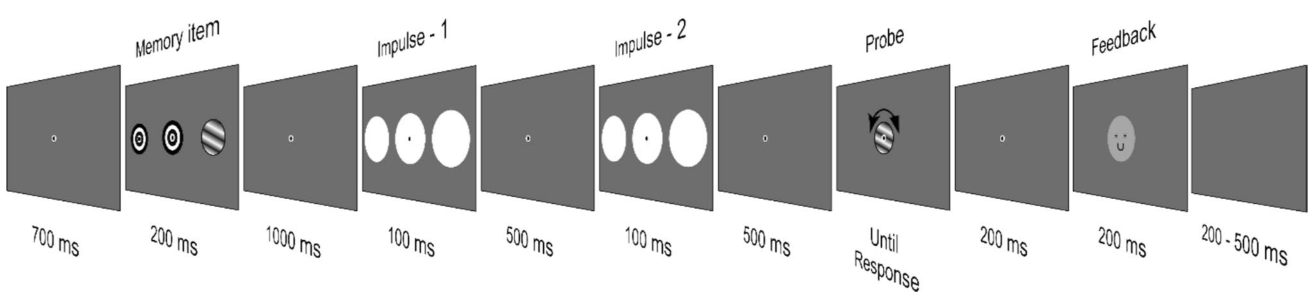

Fig. 1 Depiction of a single trial-flow. *Note*. The memory item consisted of a single sinusoidal stimulus and was displayed on one of three possible locations, whereas placeholders filled the remaining two locations. The task was to maintain the orientation of the memory item throughout impulse/distractor signals and report it by rotating the probe with the help of a joystick

volunteered in return for monetary reward (13 euros/hr) or study credits. The sample size was estimated for the EEG analyses (reported elsewhere), and based on earlier studies with similar methods (e.g., Wolff et al., 2017). Verbal and written information about the general aim and the general procedure was provided. A written consent was provided by the participants, declaring their compliance with general data collection and management procedures. Twenty-nine out of 30 participants completed all 2,880 trials whereas for one participant some technical problems resulted in the loss of 96 trials. The experiment complied with protocol VCWE-2021–173 as approved by the Scientific and Ethics Review Board of the Faculty of Behavioural and Movement Sciences, VU University Amsterdam.

Stimuli and apparatus

The background was grey (RGB = 128, 128, 128; 76.3 cd/m^2) throughout the experiment and a fixation dot (a 0.25 dva diameter white disk surrounded by a 0.3 dva diameter white circle) was always present at the origin of the screen. The memory item was an oriented sinusoidal grating (5 dva, 81.2 cd/m^2), displayed on the left, right or at the center of the screen. The item was positioned either centrally, or at 15 dva away from the center in the horizontal (either left or right) direction. The two other positions were then occupied by bullseye placeholders, consisting of concentric black and white circles. The orientation of the sinusoidal grating was randomly selected from a set of 24 discrete equidistant angles ranging from 3.75° to 176.25° (7.5° angular distance). The contrast was at 100% and the spatial frequency of the grating was kept at 1 cycle/dva, whereas the phase was a random value that could range from 1 to 180 on each trial. Each trial also contained two impulse stimuli that were irrelevant for the participant's task and for the current analyses, but were integral for the EEG experiment. These impulse stimuli consisted of three white discs (7.5 dva in diameter, RGB = 255, 255, 255; 306.4 cd/m^2) centering the same locations as the memory item and the placeholders.

At test, participants reported the memorized orientation by rotating a randomly oriented sinusoidal grating stimulus (5 dva in diameter, 81.2 cd/m^2), presented at the center of the screen. The contrast and the spatial frequency of the probe were identical to the memory item and the phase was randomly drawn on each trial. A feedback consisting of a happy (Absolute error ≤ 20°) or sad (Absolute error > 20°) faced smiley (5 dva in diameter, RGB = 175, 175, 175) was presented after response.

The stimulus generation, presentation and the experimental flow were controlled by the Psychtoolbox extension for MATLAB (Brainard, 1997; Kleiner et al., 2007). The stimuli were displayed on a 23.8-in. (60,452 cm on diagonal) ASUS ROG Strix XG248Q monitor with a resolution 1,920 × 1,080 pixels and a refresh rate of 240 Hz. The luminance output of the monitor was not linearized. A Windows compatible Xbox controller with a USB cable was used for response collection, where the left joystick could be rotated to manipulate the probe and the 'A' button was used to submit the response. The eye data was recorded by a desktop mounted EyeLink 1000 Plus (1000 Hz sampling rate) and 1024 Hz. EEG was recorded from 64 channels via BioSemi Active 2 system throughout the experiment.

Procedure

In a dimly lit room, participants were seated 60 cm away from a screen with their head placed on a chin rest. Each participant first completed a practice session consisting of 72 trials (three blocks) to learn the task, joystick usage as well as when to blink to reduce artefacts in EEG signal. Next, all participants (unless stated otherwise) completed 2880 trials, divided into five consecutive sessions over the course of 6–7 h, all completed on the same day. A single trial set-up is illustrated in Fig. 1. Participants were instructed to stay as still as possible and maintain their gaze fixated on the dot at all times. Before each session, and when the head was moved, between the blocks, the eye-tracker was calibrated using a five-dot calibration method. Between sessions it was possible to take self-regulated breaks, whereas within a session participants could take brief breaks between blocks in every 24 trials.

Participants initiated each block by pressing the SPACE bar, followed by the presentation of a 'Get Ready' message in Arial font in size 12 (400 ms). A blank screen was displayed next (600 ms), before the onset of the fixation dot. The fixation dot was displayed for the remainder of the trial. The memory item and the placeholders (200 ms) were presented 700 ms after the onset of the fixation dot, followed by a delay (1,000 ms). Next, the first impulse/distractor was presented (100 ms), followed by a delay (1,000 ms). The second impulse was next (100 ms), followed by another delay (500 ms) until the response screen was displayed. The response screen stayed on until participants completed rotating the probe to match their memory and pressed the button to submit their response. Afterwards a blank screen (200 ms) was presented, followed by the feedback (200 ms). The next trial automatically began following a random jitter (200 – 500 ms).

Data processing and analyses

The precision of responses constituted the measure of behavioural performance. The distribution of errors (deviation from memory stimulus) was assessed by fitting a von Mises distribution using 'circ_vmpar' function of the Circular Statistics Toolbox (Berens, 2009). Estimated k values (inverse variance of fitted von Mises distribution) increased with higher precision. *K* values at different eccentricity were contrasted with two-tailed pairwise *t* tests, reflecting performance differences across the visual field. These results are presented in detail in Supplement–1 (Fig. S1).

The trial-wise error was calculated as the difference of the reported orientation from the real orientation of the presented memory item. For each presentation location, trials with errors 3 standard deviations and more away from the mean were excluded as outliers. This corresponded to less than 1.7% of all trials (Mean = 46.38, Std. = 24.9, with $M_{\text{periphery}} = 35.29$, Std. = 19.99, and $M_{\text{center}} = 11.09$, Std. = 8.47). Furthermore, the first trials of all blocks were excluded from the analysis as well due to the break between blocks. The error values of the remaining trials were grouped as a function of the difference of the memory item from the previous trial orientation. To avoid individual biases and their effects on error distributions (Barbosa & Compte, 2020), error by relative orientation distribution was folded across the origin. For this, the sign of the error values in trials with negative relative orientation ($error_{(n-1)}$—$error_{(n)} < 0$) were reversed. The folded trials were then grouped as a function of absolute relative orientation and the presentation locations of the current and preceding trial. The grouped trials were averaged, forming a single curve per condition for each participant. In total, each participant provided 5 folded curves; corresponding to when consecutive trials were presented at the same location (*peripheral* → **PERIPHERAL (same)**; *central* → **CENTRAL**), when locations switched from peripheral to central (*peripheral* → **CENTRAL**) and central to peripheral visual field (*central* → **PERIPHERAL**), and when location switched across peripheral locations (*peripheral* → **PERIPHERAL (switch)**). Furthermore, differences between these curves were calculated by subtracting mean errors at each relative orientation.

A cluster-corrected permutation test was applied to test whether curves and the differences between these curves deviated from zero for relative orientation. The cluster test treats subject-wise curves as series of correlated data (as is also the case for example in time series) and ensures that the observed effect is supported by multiple bins. This approach prevents false positives stemming from chance inflation by multiple comparisons based on single bins. For this, a null distribution was formed by flipping the sign of mean error values 100,000 times with 50% chance at each relative orientation. The group mean of all participants was contrasted against this null-distribution with the proportion corresponding to the probability of observing the original mean error cluster reported as the *p* value (cut-off $p < 0.05$). These tests were two-tailed.

Considering that serial dependence effects tend to peak for orientation differences of ~ 22.5° (Manassi et al., 2023, meta-analysis arrived at 25% of 90°), we also separately analysed the mean error when memory items differed between 15° to 30° from the previous trial. The statistical significance of the mean errors for distinct item locations in the current and previous trials were assessed with repeated-measures analyses of variance (RM-ANOVAs), using the freely available statistics program JASP (JASP Team, 2024).

Additionally, we assessed serial bias of the stimulus in the current trial towards the response in the preceding trial. Data from each participant was first filtered for outliers, using the same criteria explained earlier. Next, we grouped trials based on the difference between previous response and the stimulus in the current trial. These trials were then binned according to discrete stimulus–stimulus differences to allow comparison of bias relative to previous stimulus. Next, the sign of the errors in trials with negative differences were flipped to collapse data over absolute differences and errors were averaged, yielding a single curve for each participant in each eccentricity condition. The remainder of the analyses was identical to investigation of bias to preceding stimulus, as described earlier. These results are presented in supplements (Supplement–2, Fig. S2 and S3).

Results and discussion

Performance overall was high throughout the task (see Fig. S1A in Supplement–1 for probability density function of errors). Responses were more precise for central stimuli as reflected by larger *k* values estimated in this condition (Fig. S3D), $t(29) = 8.669$, $p < .001$. Reduced performance in the periphery was aligned with the diminishing effects of eccentricity on vision.

Next, we assessed the presence of serial dependence across central and peripheral presentation locations. Figure 2 shows the folded error curves as a function of the absolute orientation difference, where positive deviations from zero indicate a bias towards the orientation on the previous trial. Cluster-corrected permutation tests against zero revealed that the previous trial had a significant impact on the current trial in all conditions—namely, when the current item was at the center preceded by an item at the same location (Fig. 2A, *central* → **CENTRAL,** dark green solid line, range: 7.5°–52.5° difference, $p < .001$) or preceded by an item in the periphery (*peripheral* → **CENTRAL** light green solid line, range: 7.5°–52.5° difference, $p < .001$); and when the current orientation was in the periphery either preceded by another peripheral orientation at the same location (Fig. 2B, *peripheral* → **PERIPHERAL (same),** dark-blue solid line, range: 7.5°–52.5°, $p < .001$), or by a central item (*central* → **PERIPHERAL**, light blue solid line, range: 7.5°–45° difference, $p < .001$). Even when both consecutive items were presented in the periphery but at opposite ends, there was a clear serial bias (Fig. 2C, *peripheral* → **PERIPHERAL (switch)**, burgundy solid line, range: 7.5°–52.5° difference,

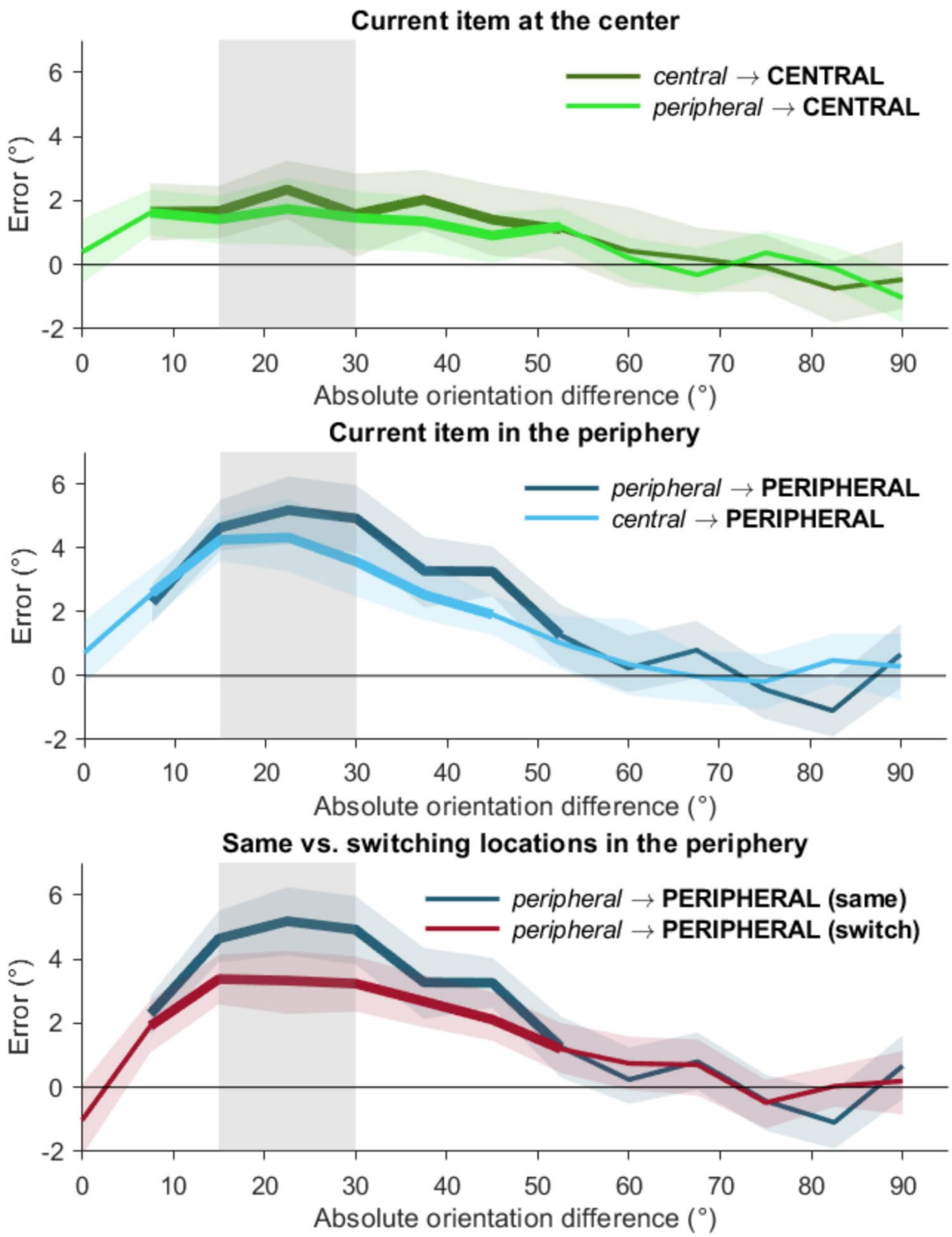

Fig. 2 Serial dependence at periphery and center in Experiment 1 when current item is presented at center (top), periphery (middle) or when consecutive items at periphery appear at the same or different locations (bottom). *Note*. Errors for orientations presented at distinct locations with specific preceding orientation location as a function of absolute orientation difference between trials, folded across the origin. Positive deviations from zero indicate attraction towards the preceding trial. Error curves reflect mean error for when the current orientation is at center (top), preceded by an orientation in the periphery (light green), or at center (dark green); when the current orientation is in the periphery (middle), preceded by an item at the center (light blue) or in the periphery (dark blue); and when both consecutive orientations were in the periphery (bottom), at the exact same location (dark blue) or at distinct peripheral locations (burgundy). The thicker line segments mark the absolute orientation difference range within which the serial dependence is significantly different from 0 according to cluster corrected permutation test ($p < .05$). The shading around the lines mark 95% C.I. The grey zone marks the range of absolute orientation difference within which the errors were averaged for contrasting serial dependence at the current and the preceding item locations. (Colour figure online)

$p < .001$). Thus, in line with previous studies we found that responses were pulled towards the preceding orientation, with the attraction predominantly occurring between 7.5° to 45° orientation differences (Fischer & Whitney, 2014; Manassi et al., 2023; Pascucci et al., 2023). We also analysed the biases in relation to the previous response rather than the previous stimulus (see Supplement–2). Here the attraction was somewhat stronger consistent with earlier reports (e.g., Pascucci et al., 2019). The overall pattern remained the same though, as here too the bias was stronger for peripheral than for central stimuli (see Supplement–2, Figs. S2 and S3). We therefore focus our report on the stimulus-based analyses.

Figure 3 shows these biases averaged across the range of 15°–30° absolute orientation difference. This allowed us to directly compare effects of eccentricity in the current trial to the effects of eccentricity of the preceding trial. We conducted an RM-ANOVA on the data of the first four columns of Fig. 3, with current trial location (central, peripheral) and previous trial location (central, peripheral) as factors. This only yielded a significant main effect for the current trial location, $F(1,29) = 82.026$, $MSE = 2.416$, $p < .001$, $\eta^2 = 0.517$, with the strongest bias again for peripheral orientations. This difference was observed for 29 of the 30 participants. There was no effect of the preceding trial location, $F(1,29) = 1.102$, $MSE = 1.533$, $p = .423$, and no interaction, $F(1,29) = 3.073$, $MSE = 2.159$, $p = .09$.

Previous work has shown that serial dependence declines with interstimulus distance (Collins, 2019; Fischer & Whitney, 2014; Manassi et al., 2023). To see if similar distance effects occurred here, we analysed the three right-most conditions as plotted in Fig. 3—namely, *peripheral* → **PERIPHERAL (same;** dark blue), *central* → **PERIPHERAL** (light blue), and *peripheral* → **PERIPHERAL (switch;** burgundy). These conditions represent an increasing distance between previous and current item location while keeping current item location the same. A one-way RM-ANOVA with these three conditions as levels revealed a main effect of distance, $F(2,58) = 10.048$, $p < .001$, $\eta^2 = 0.257$. Individual paired *t* tests revealed a significant decline between *peripheral* → **PERIPHERAL (same)** and *central* → **PERIPHERAL**, $t(29) = 2.441$, $p = .021$, as well as between *central* → **PERIPHERAL** and *peripheral* → **PERIPHERAL (switch),** $t(29) = 2.055$, $p = .049$. Consequently, the difference between *peripheral* → **PERIPHERAL (same)** and *peripheral* → **PERIPHERAL (switch)** was also significant, $t(29) = 4.445$, $p = .001$. Of further noteworthiness, an additional comparison showed that the serial dependence in the *peripheral* → **PERIPHERAL (switch)** condition was still stronger than in the *peripheral* → **CENTRAL,** $t(29) = 6.234$, $p < .001$, and the *central* → **CENTRAL** conditions, $t(29) = 4.261$, $p < .001$, despite the latter combinations reflecting shorter distances. This only further confirms that peripheral stimuli are subject to stronger biases than central stimuli.

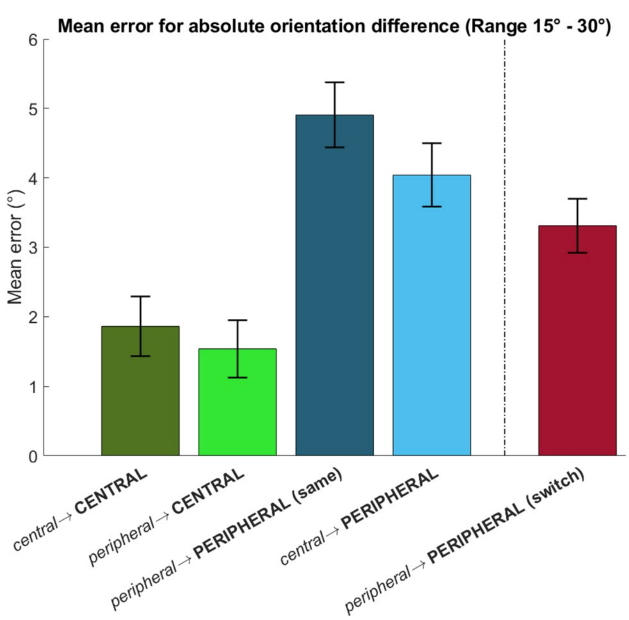

Fig. 3 Mean error within the range of 15° to 30° absolute orientation difference in Experiment 1, averaged for the different current and preceding item location combinations. *Note*. Mean error for absolute orientations differences within the range of 15° to 30° in consecutive trials, grouped for each current and preceding item location. The whiskers mark the 95% C.I. within-participant error. (Colour figure online)

Discussion

In line with previous studies, we found that responses were attracted towards the preceding orientation (with the attraction predominantly occurring between 7.5° to 45° orientation differences), and that this serial dependence effect generally declined with distance between stimuli (Collins, 2019; Fischer & Whitney, 2014; Manassi et al., 2023; Pascucci et al., 2023). The important novel finding is that the serial dependence was not uniform across the visual field, as the attraction bias was stronger for peripheral than for central patterns. This is consistent with the starting hypothesis that peripheral vision is subject to stronger overlap and therefore stronger similarity/continuity of signals, and at the same time reduced precision and increased uncertainty (as also indicated by overall performance levels in the periphery), both factors that have been thought to promote integration between current and preceding percepts. We furthermore observed that it was only the eccentricity of the *current* trial that determined the extent of the serial dependence. This aligns with previous studies showing the level of current trial uncertainty as the main determinant of serial dependence (Braun et al., 2018; Ceylan et al., 2021; Gallagher & Benton, 2022; Samaha et al., 2019; Suárez-Pinilla et al., 2018).

Experiment 2

The main finding of Experiment 1 was a stronger bias towards the preceding item in the periphery relative to central vision. We attribute this to the lower fidelity that comes with peripheral vision, resulting in relatively stronger integration with, and reliance on previous perceptual evidence. However, attentional factors may also have played a role. In Experiment 1, participants did not know at the start of a trial where the pattern would appear—at the left, right, or the center. This may have had multiple implications in terms of attention. For one, as attention is typically concentrated around the center of vision (Donk et al., 2024; LaBerge, 1983; LaBerge & Brown, 1986; Olivers et al., in press; Wolfe et al., 1998; see Olivers, 2025, for a comprehensive review), and as a result, peripheral patterns may not only have suffered in terms of the feedforward signal, but also have received less attention than central patterns. As the lack of attention may have affected perceptual precision further (Yeshurun & Carrasco, 1998), there may have been more room for serial dependence effects to emerge for peripheral patterns. However, the converse may also be the case: Previous work has indicated that serial dependence effects actually become *stronger* for attended stimuli (Manassi et al., 2023; Whitney & Fischer, 2014). If attention was indeed biased towards the center, then this would only lead to an underestimation of the differences in serial dependence between central and peripheral stimuli. Finally, note that in Experiment 1, combining the left and right eccentric items, there were twice as many peripheral targets than central targets. This raises again another possibility—namely, that observers decided to a priori attend *more* to eccentric than to central items. There is some evidence that observers can split their attention, or potentially even attend to a ring-like region (Awh & Pashler, 2000; Castiello & Umiltà, 1992; Egly & Homa, 1984; Linnell & Humphreys, 2004). Given the aforementioned findings—that attention enhances serial dependence effects—the stronger such effects for peripheral items may have been caused by them having received more, rather than less, attention. To assess whether the observed effects in Experiment 1 were due to differences in central versus peripheral vision, or merely caused by attention, Experiment 2 sought to equate attention by adding a cue that was informative (valid) on the location of the upcoming target pattern. For comparison, we also added noninformative (neutral) cues, serving as a control. We hypothesized that if the eccentricity-based serial dependence differences as observed in Experiment 1 were due to attentional differences, then informative cues should result in a clear reduction in such differences, compared with the neutral cue baseline.

Methods

Participants

The participant sample consisted of 30 healthy adults ($M_{\text{Age}} = 23.83$ years, $SD_{\text{Age}} = 9.07$, range: 18–63, 26 women), who participated in this study in return for a monetary reward (10 euros/hr) or study credits. The sample size was kept the same as Experiment 1. Participants received information on general data collection and management procedures and provided written consent. The protocol (VCWE-2021–173) was approved by the Scientific and Ethics Review Board of the Faculty of Behavioural and Movement Sciences, VU University Amsterdam.

Stimuli and procedure

Figure 4 presents the trial design for Experiment 2, which was identical to Experiment 1, except for the following. The novel additional manipulation in Experiment 2 concerned the presence of a cue. The cue stimuli consisted of two diamond shapes, sized ~ 1.3 dva and centered 10 dva to the left and right from the center, between the central and peripheral placeholders. The diamonds were split into a blue (RGB = 0, 0, 128) and a red (RGB = 128, 0, 0) half, thus forming two arrow heads that, depending on which cue colour was relevant in a certain session would either cue inwards towards the central item (central cue), to the left or right for a peripheral item (peripheral cue), or up/down (neutral cue). The relevant cue colour was communicated before each session and the colour-order was counterbalanced across participants.

Unlike in Experiment 1, there were no impulse signals and the delay period was therefore reduced to 1,800 ms. Furthermore, three placeholders consisting of black empty circles were displayed on the screen throughout a trial (except probe and feedback display). The purpose of the placeholders was to increase the utility of the cue so that observers could focus on the anticipated stimulus location. The placeholders were of the same size as the memory items.

On each trial, first the fixation dot and placeholders were presented for 200 ms. Then the cue was displayed for 200 ms, followed by a delay with a duration of 1,000 ms. Next the memory item and two fillers (bullseye patterns) were displayed for 200 ms. After a delay of 1,800 ms, the probe was presented. Participants could rotate this probe with the joystick to report the memory item and press ‘A’ to submit their answer. Following a 200-ms blank screen, a feedback screen consisting of a sad/smiling smiley was displayed for 200 ms. A random delay (jittered between 200 and 500 ms) preceded the next trial.

All participants completed 48 practice trials with the specific colour-cue before each session. Practice trials were identical to test trials. In total 29 subjects completed 1,152 test trials, divided into two sessions, over a course of 2.5 h whereas one subject completed 1,008 trials due to a technical error that stopped the experiment prematurely. All practice and test trials were completed on the same day by each participant.

Behavioural analyses

Overall performance As in Experiment 1, we estimated the k values of a fitted von Mises distribution to errors for each

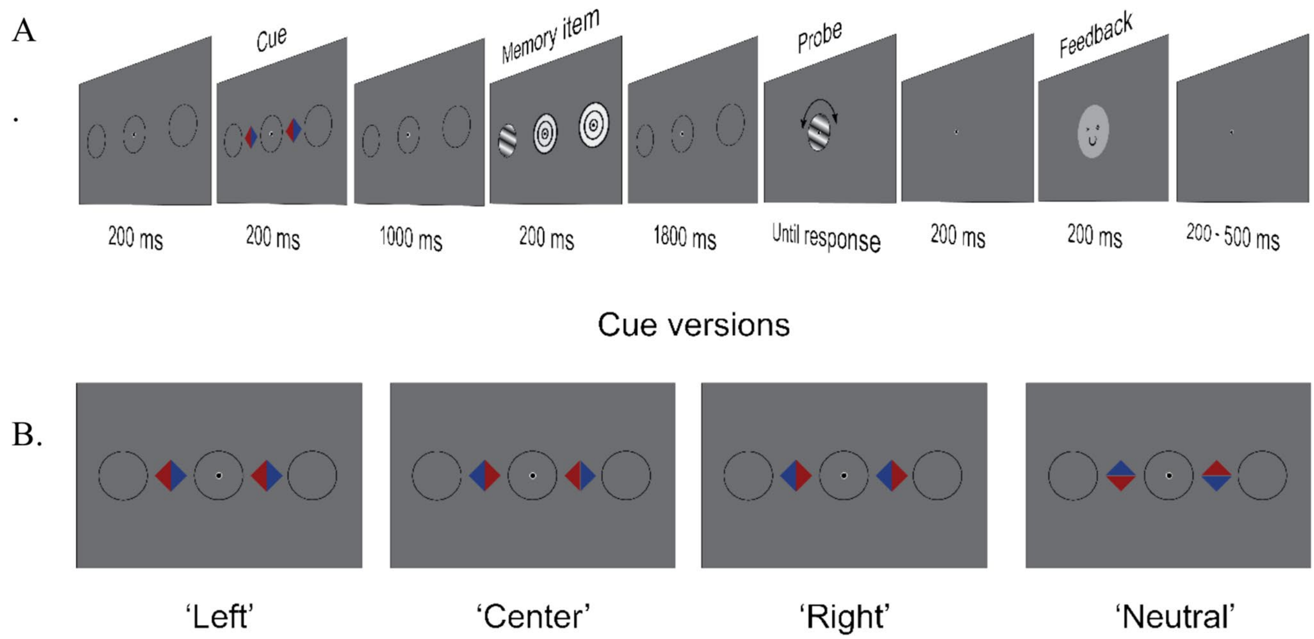

Fig. 4 The design of a single trial (**A**) and possible cues (here for red as the relevant colour). *Note*. A single memory item consisting of a sinusoidal grating with an orientation was presented either on the left, right or center and task is to memorize and report the orientation. The other possible presentation locations were filled by concentric black and white circles serving as placeholders. In half of the trials a valid cue, preceding the memory item, addressed the location of the orientation-to-be-memorized whereas in neutral cue trials the cues did not point at a location in coherence, where the memory item could be presented. Participants were required to maintain the orientation and report it by rotating the probe with the help of a joystick. The possible cue displays are depicted (**B**) for a session in which the cue colour was red when it was valid (the three examples on left) and neutral (Right). (Colour figure online)

participant, as a measure of precision. Here we sought to assess whether cue manipulation influenced performance and if so, whether this effect interacted with location. The estimated *k* values were tested with omnibus RM-ANOVA on freely available statistics program, JASP (JASP Team, 2024).

Serial dependence In Experiment 2, we assessed folded error values for the absolute orientation difference relative to the previous trial for each current trial location. Since Experiment 1 revealed that current item location (central vs. peripheral) and not the preceding item location determined the strength of the serial dependence, we collapsed across the latter. We grouped data in two ways: First, we grouped errors based on cue condition in the current trial and contrasted serial dependence across peripheral and central locations. Second, we re-did this grouping based on the preceding trial's cue condition. For the rest, data preparation, averaging, and statistical analyses were identical to Experiment 1.

To align with the analyses of Experiment 1, errors were also re-analysed in relation to the location of memory item in the preceding and current trial. Due to reduced trial count, for this analysis the data were collapsed across cue conditions. The results are reported in Supplement–3 with curves presented in Fig. S4.

Results and discussion

Cueing effects on memory performance The probability density functions for error distributions across all cue and eccentricity conditions are presented in Supplement–1 (see Fig. S1B). We contrasted estimated precision (*k* values) of the von Mises distribution fitted to errors in each condition for each participant for each location (central, peripheral) and cueing condition (cued, neutral), as presented in Fig. S1E. The RM-ANOVA on k values ($M_{\text{cued \& peripheral}}$ = 4.631, $M_{\text{uncued \& peripheral}}$ = 4.606, $M_{cued\ \&\ central} = 7.750$, $M_{uncued\ \&\ central} = 7.121$) revealed a location, $F(1,29) = 49.302$, $MSE = 4.829$, $p < .001$, $\eta^2 = 0.570$, and a cue main effect, $F(1,29) = 5.702$, $MSE = 0.562$, $p = .024$, $\eta^2 = 0.008$. The interaction also reached significance, $F(1,29) = 4.487$, $MSE = 0.562$, $p = .043$, $\eta^2 = 0.007$, indicating that it was mainly central orientations which benefitted from the cue. Thus, performance benefitted from the cue, indicating that observers did make use of the information provided. The benefit being largest for the central pattern, however, is somewhat unexpected. This may reflect the fact that on two-thirds of the trials, the target would appear in a peripheral location, and so observers were already biased towards those locations. In fact, in hindsight, this may have contributed to a larger serial dependence effect in Experiment 1, and is something we will further address in Experiment 3. If so, then the current cueing manipulation appears to have contributed to correcting this bias.

Serial dependence Figure 5A shows the error as a function of the absolute orientation difference, where deviations from zero in positive direction reflect attraction towards the previous trial value. At first glance it is noticeable that the biases were overall smaller than Experiment 1, which could be related to the differences between the experiments (cueing, shorter delay periods, fewer trials; cf. Barbosa et al., 2020; Bliss et al., 2017; Ceylan & Pascucci, 2023).

The analysis of mean error as a function of absolute orientation difference in reflected by folded curves, replicated Experiment 1 and revealed a bias towards the previous trial both for peripheral (Fig. 5A, blue solid line, range: 7.5°–60°, $p<.001$) and central items (Fig. 5A, green solid line, range: 15°–30°, $p<.001$). The serial dependence in the periphery was also again significantly larger than at the center

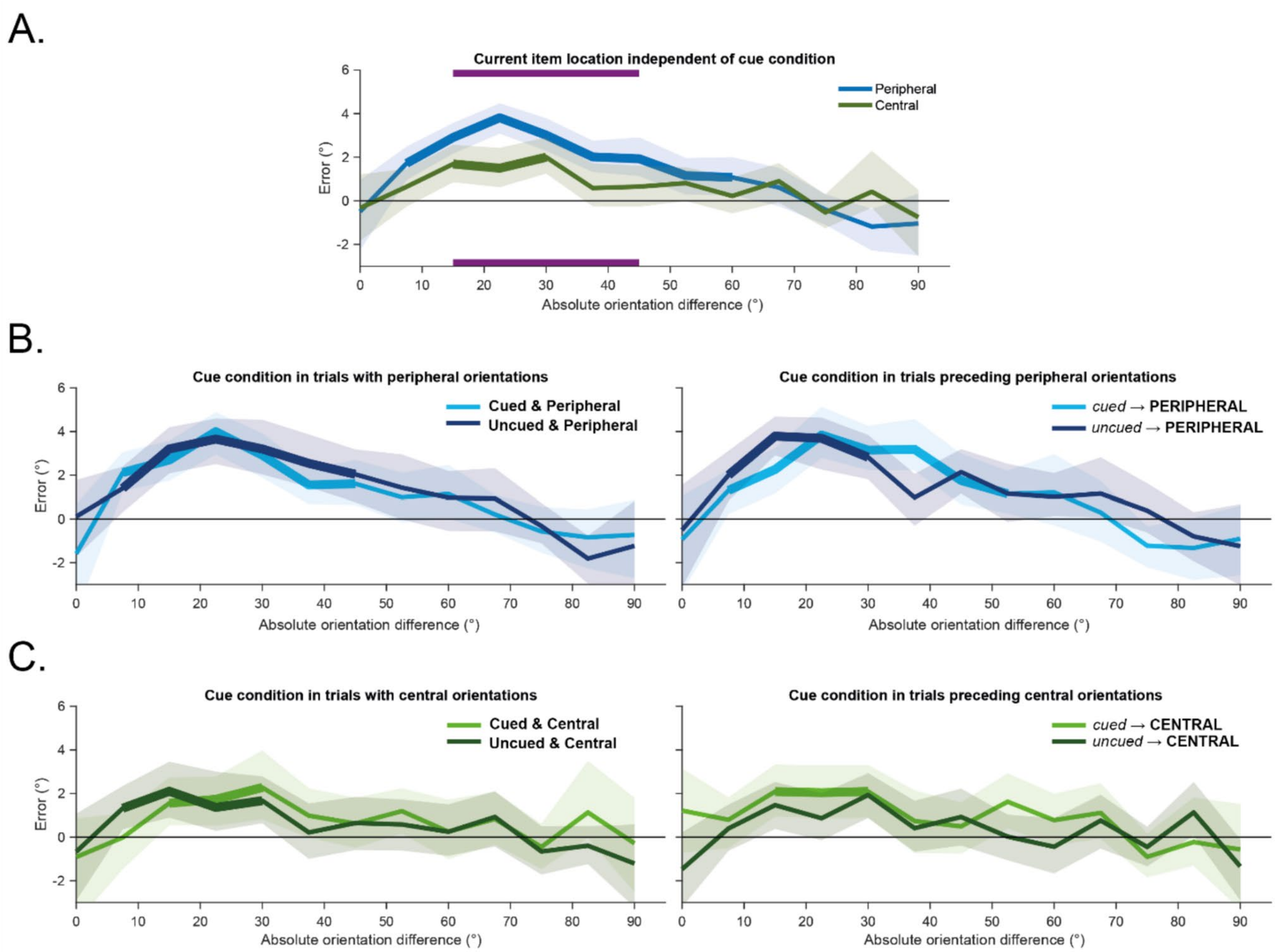

Fig. 5 Serial dependence at periphery and center in Experiment 2 collapsed across cue conditions (**A**), at periphery when current (left) or preceding (right) trial is cued (**B**), and at center when current (left) and preceding (right) trial is cued (**C**). *Note*. Error as a function of the absolute orientation difference, folded across the origin. Positive deviations from zero indicate attraction towards the preceding trial. (**A**) Errors for orientations in the periphery (blue) and center (green), collapsed across cue condition. Solid purple bars at the top and bottom mark the range of absolute orientation difference at which serial dependence differs significantly for peripheral and central items ($p<0.05$). (**B**) Errors presented separately for cued (light blue) and uncued (dark blue) current trials (on the left), and cued and uncued previous trials (on the right) when current orientation is in the periphery. (**C**) Errors for cued (light blue) and uncued (dark green) current trials (on the left), as well as cued and uncued previous trials (on the right), when the current item was presented at the center. Solid lines show mean error whereas thicker line segments mark the range of significant serial dependence as a function of absolute orientation difference according to cluster corrected permutation test ($p<0.05$). The shading around the lines mark 95% C.I. (Colour figure online)

(Fig. 5A, purple zone, range: 15°–45°, $p < .001$). As can be observed from Figs. 5B and C, respectively, no cueing-related differences for neither peripheral nor central items were apparent, regardless of whether the items were cued on the current trial or on the previous trial.

Figure 6 shows the average serial dependence effect within the 15–30° range, as a function of current (left panel) or previous trial (right panel) cueing condition. As before, we observed a stronger serial dependence for peripheral items than for central items, $F(1,29) = 16.531$, $MSE = 4.051$, $p < .001$, $\eta^2 = 0.204$. This difference was observed for 23 of the 30 participants. However, there were no statistically reliable effects of current cue condition, $F(1,29) = 0.002$, $MSE = 2.500$, $p = .988$, nor of preceding cue condition, $F(1,29) = 0.353$, $MSE = 1.919$, $p = .557$, nor any interactions with cueing, all $F\text{s} > 2.3$, $p\text{s} > .137$. Although recall performance was slightly improved by the precue, our manipulation of attention did neither influence the induction of serial bias, nor the attraction observed in the current trial.

Discussion

The results of Experiment 2 replicated those of Experiment 1 and revealed stronger serial dependence in the periphery. Importantly, cueing had little effect on serial dependence, irrespective of the location of the item in the current trial. This suggests that attention is not a major factor in explaining the differences in serial dependence between central and peripheral vision. Note that overall performance was still influenced by the cue, mainly for central orientations. This is consistent with the idea that attention may have been initially biased towards peripheral items, potentially because two-thirds of the trials carried a peripheral target. This is something we will further address in Experiment 3. The fact that cueing led to performance improvement for central items while the strength of serial dependence remained the same, further points to attention playing little role in the current findings. This supports the conjecture that serial dependence largely reflects automatically generated biases (Cicchini et al., 2024).

Experiment 3

Experiment 3 sought to eliminate any potential confounding resulting from the uneven distribution of peripheral versus central items. In Experiment 3, on 50% of trials the target item was presented at the center, whereas on the remaining 50% it was presented in the periphery (25% left, 25% right). Furthermore, based on a reviewer's suggestion, we also adjusted the contrast levels of the gratings from the relatively high contrast used in Experiments 1 and 2, to the more commonly used lower contrasts used in previous serial dependence studies (e.g., Ceylan et al., 2021; Fischer & Whitney, 2014).

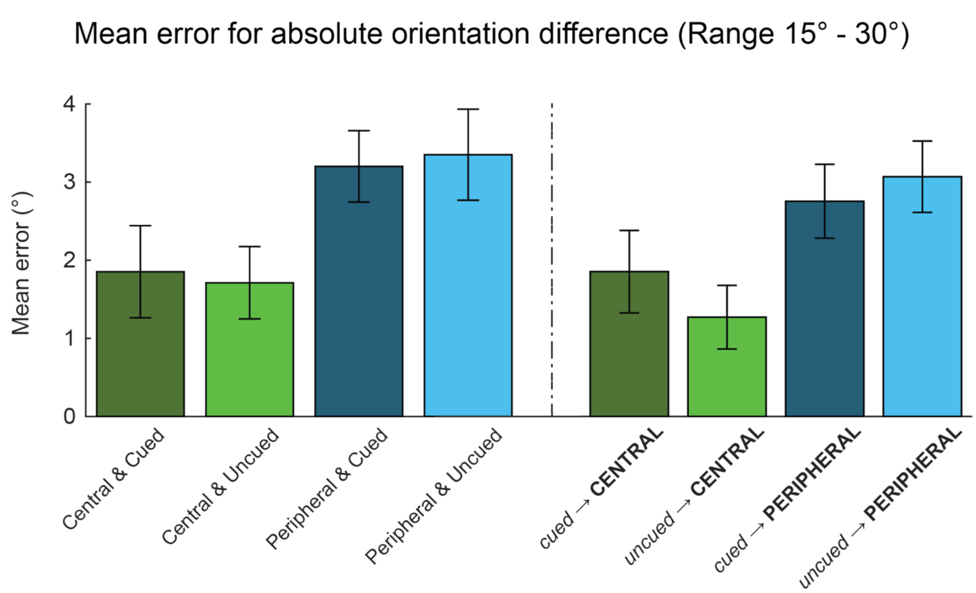

Fig. 6 Mean error within the range of 15° to 30° absolute orientation difference in Experiment 2 for peripherally and centrally presented orientations when cue condition in current trial is concerned (left) and when cue condition in preceding trial is concerned (right). *Note*. The bar graphs depict mean error within the range of 15° to 30° absolute orientation difference as a for distinct item location in the current trial grouped in accordance with the cue condition in the current (left) or preceding trial (right). The whiskers mark the 95% C.I. within subjects. (Colour figure online)

Methods

Participants

The participant sample consisted of 30 healthy, young adults ($M_{\mathrm{Age}} = 21.40$, $SD_{\mathrm{Age}} = 3.81$, range: 19–39, 24 women) who volunteered to take part in this study in return for monetary reward (10 euros/hr) or study credits. The sample size was chosen to be the same as Experiments 1 and 2. Participants were informed about experimental procedures and data management protocol both verbally and in written form and provided written consent showing their compliance with the protocol. The Scientific and Ethics Review Board of the Faculty of Behavioural and Movement Sciences, VU University Amsterdam approved the protocol (VCWE-2021–173).

Stimuli and procedure

Stimuli used in Experiment 3 were identical to Experiments 1 and 2, except that we reduced the contrast level for memory items and place holders by 50%, we removed the cueing manipulation, and reduced the probability of an item appearing in the periphery was to 50% (25% left, 25% right; with the remaining 50% appearing in the center). A typical trial is illustrated in Fig. 7.

Each trial began with the presentation of the fixation dot for 700 ms. This was followed by the presentation of a disc (5 dva in diameter) containing a sinusoidal grating with an orientation (one of 24 equidistant orientations ranging from 3.75° to 176.25° with 7.5° in between) at the center (50% probability) or 15 dva lateral to the fixation dot in the periphery (25% of which were on the left and 25% on the right side). All other possible presentation locations were occupied by placeholders formed by concentric black and white overlapping circles with no orientation information. The contrast of all stimuli was 0.5 whereas the spatial frequency was 1. The memory item was presented for 200 ms, followed by a delay with 1,800 ms duration. Next a single sinusoidal grating with a random orientation was presented at the center. Participants could rotate this orientation with the help of the joystick on an Xbox controller compatible with the PC and submitted their responses by pressing A on the same joystick. After response submission, 100-ms delay with no visual presentation followed. Next, a feedback consisting of a sad or smiling smiley was presented for 200 ms, indicating that the response was correct (error ≤ 20°) or incorrect (error > 20°). Participants completed 1,152 trials divided over 48 blocks in two sessions which were completed on the same day. Like Experiments 1 and 2, simultaneous gaze position was monitored and recorded by a desktop mounted EyeLink 1000 Plus at 1000 Hz sampling rate. Each subject completed 24 practice trials and 1,152 test trials, over a course of 2.5 h, all on the same day.

Behavioural analyses

The analyses for overall performance as well as serial dependence was identical to Experiment 1.

Results and discussion

Participants again performed the task with relative ease (see Fig. S1C in Supplement–1 for probability density function of errors). Like the previous two experiments, performance was better when the memory item was presented at the center. Estimated precision (K) values for the centrally presented orientations were significantly larger than for the ones in the periphery (Fig. S3E), $t(29) = 8.619$, $p < .001$.

Folded error curves for Experiment 3 are presented in Fig. 7. As in Experiment 1, an attraction bias towards the previous orientation was observable for central items when preceded by an item at the center (Fig. 8A, *central* → **CENTRAL,** dark-green solid line, range 7.5°–30° difference, $p < .001$) or an item in the

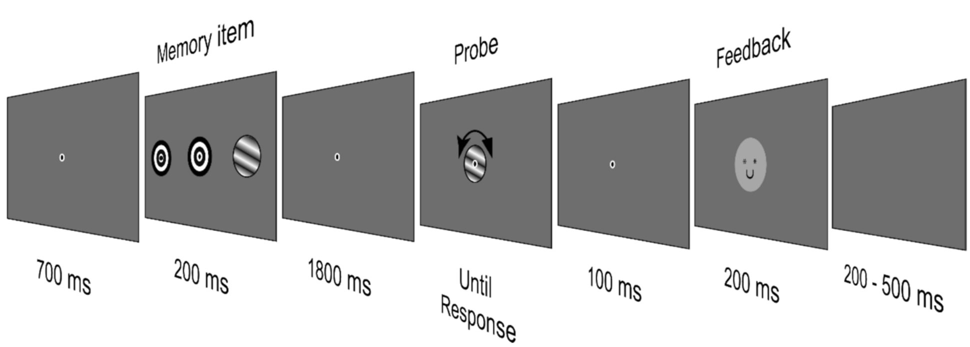

Fig. 7 Depiction of a single trial-flow in Experiment 3. *Note*. The memory item consisted of a single sinusoidal grating that could appear at the center with 0.5 probability and on the left or right side with 0.25. The task was to maintain the orientation of the memory item until a probe with a random orientation was presented. The probed grating could be rotated with the help of a joystick to re-generate the memorized orientation. The response was submitted by pressing another button on the joystick

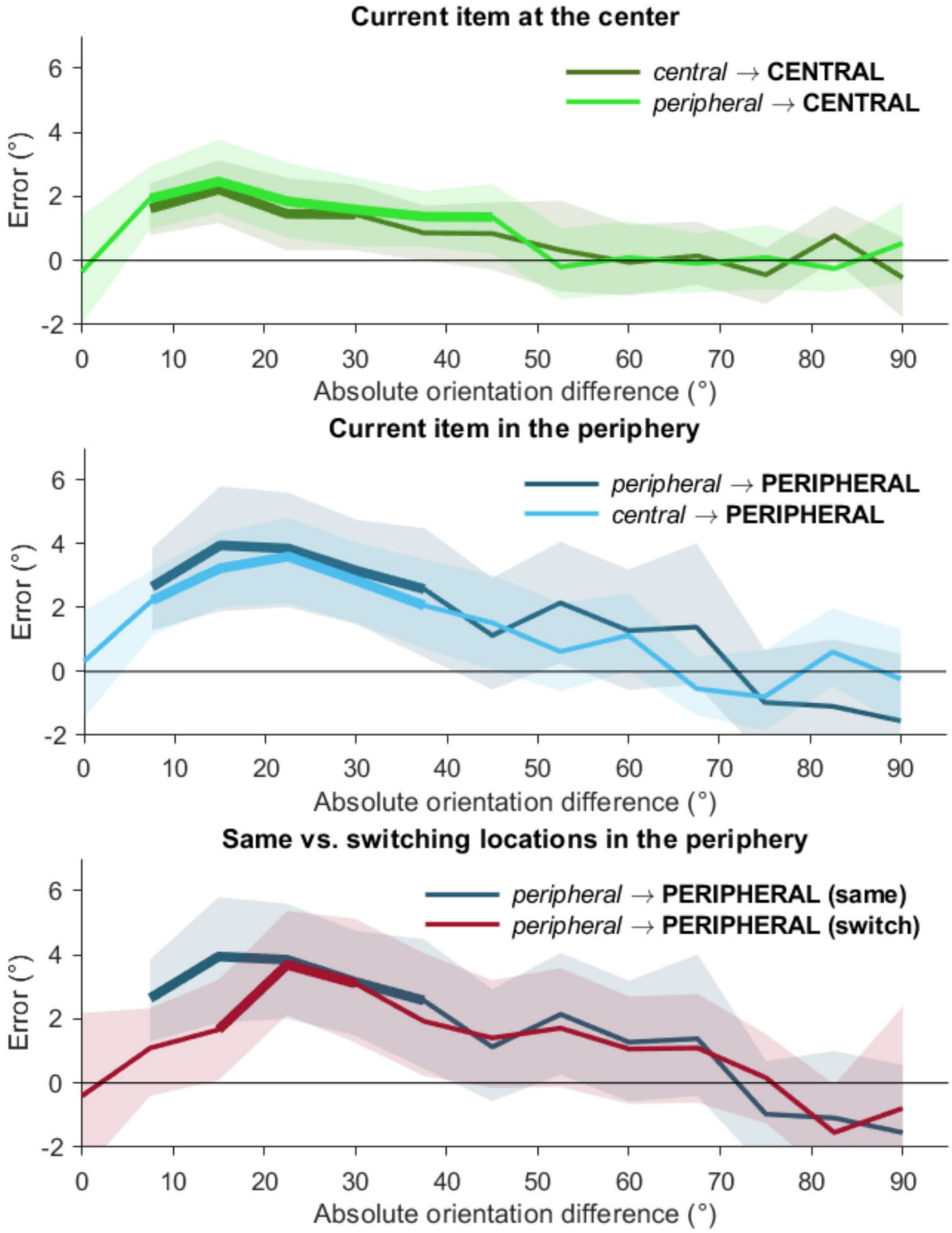

Fig. 8 Serial dependence at periphery and center in Experiment 3 when current item is presented at center (**A**), periphery (**B**), or when consecutive items are in the periphery appear at the same or different locations (**C**). *Note*. Errors for orientations presented at distinct locations with specific preceding orientation location as a function of absolute orientation difference between trials, folded across the origin. Positive deviations from zero indicate attraction towards the preceding trial. Error curves reflect mean error for when the current orientation is at center (**A**), preceded by an orientation in the periphery (light green), or at center (dark green); when the current orientation is in the periphery (**B**), preceded by an item at the center (light blue) or in the periphery (dark blue); and when both consecutive orientations were in the periphery (**C**), at the exact same location (dark blue) or at distinct peripheral locations (burgundy). The thicker line segments mark the absolute orientation difference range within which the serial dependence is significantly different from 0 according to cluster corrected permutation test ($p < 0.05$). The shadings around lines mark 95% C.I. The grey zone marks the range of absolute orientations difference within which the errors were averaged for contrasting serial dependence at the current and the preceding item locations. (Colour figure online)

periphery (*peripheral* → **CENTRAL** light-green solid line, range: 7.5°–45° difference, $p < .001$). The same was true for peripheral items, both when preceded by another peripheral item at the same location (Fig. 8B, *peripheral* → **PERIPHERAL (same),** dark-blue solid line, range: 7.5°–37.5°, $p < .001$), or by a central item (*central* → **PERIPHERAL**, light-blue solid line, range: 7.5°–37.5° difference, $p < .001$). When both consecutive

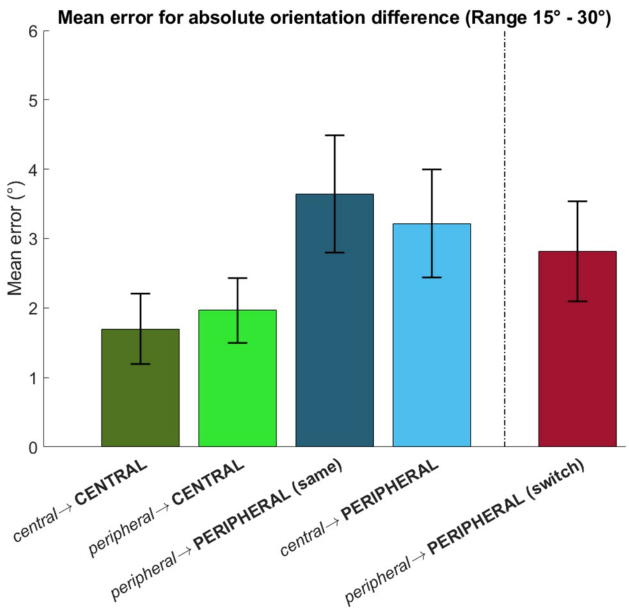

Fig. 9 Mean error within the range of 15° to 30° absolute orientation difference in Experiment 3, averaged for the different current and preceding item location combinations. *Note*. Mean error for absolute orientation differences within the range of 15° to 30° in consecutive trials, grouped for each current and preceding item location. The whiskers mark the 95% C.I. within participant error. (Colour figure online)

items were presented in the periphery at opposite ends, this serial bias was smaller in terms of amplitude and orientation range (Fig. 8C, *peripheral* → **PERIPHERAL (switch)**, burgundy solid line, range: 15°–30° difference, $p < .001$).

Figure 9 shows the serial bias, averaged across the range of 15°–30° for each condition. We tested the effect of current and previous item location (central, peripheral) with RM-ANOVA. Replicating Experiment 1, this yielded a significant main effect for only the current trial location, $F(1,29) = 21.451$, $MSE = 3.573$, $p < .001$, $\eta^2 = 0.165$, with the strongest bias again in the periphery. This difference was observed for 24 of the 30 participants. There was neither a main effect of the location of the orientation in the previous trial, $F(1,29) = 0.593$, $MSE = 6.011$, $p = .45$,, nor an interaction, $F(1,29) = 0.054$, $MSE = 3.689$, $p = 0.82$. Pairwise comparisons also again showed that the *peripheral* → **PERIPHERAL (switch)** condition still yielded a stronger bias than the *central* → **CENTRAL** condition, $t(29) = 2.198$, $p = .036$, (two-tailed), and the same was true when compared with the *peripheral* → **CENTRAL** trial pair, although not significantly so, $t(29) = 1.629$, $p = .114$ (two-tailed).

Discussion

The pattern of results of Experiment 3 replicates that of Experiment 1. Serial attraction biases were stronger for peripheral than for central items, and this effect depended on current trial location rather than previous trial location. Given that we here equalized the probability of central versus peripheral items, we can conclude that the uneven distributions in Experiments 1 and 2 were not a primary cause. We note that overall biases were less strong than in Experiment 1. Several differences may have contributed here. First, the luminance contrast of the gratings was diminished. Second, we shortened the experiment by running fewer trials, while previous work has suggested that serial dependence may get stronger over time (Barbosa et al., 2020), plus less (though still sufficient) statistical power. Third, we implemented a shorter delay before the response. Previous work has shown that serial bias tends to increase with longer memory delays (Bliss et al., 2017). Despite these differences, the main result clearly replicated: serial dependence effects were stronger for peripheral than for central stimuli.

General discussion

Perceptual judgements are not independent from past experiences, as responses can be influenced by preceding perceptions when consecutive stimuli are similar in terms of a task-relevant feature (Cicchini et al., 2014; Fischer & Whitney, 2014). We investigated how visual eccentricity influences serial dependence, given that perception is not uniform for central and peripheral vision. Across three experiments using orientation stimuli, we observed clear serial dependence, with judgements being attracted towards previous orientations. Consistent with earlier reports (see Manassi et al., 2023, for a review and meta-analysis), this bias was maximal within a 15–30° range. Importantly, this attractive bias towards the preceding trial was not uniform across the visual field. In all three experiments, we observed stronger serial dependence for orientations presented in the periphery relative to the center.

An important question is why serial dependence is stronger for peripheral vision. Theoretical accounts of serial dependence differ, but all assume that it serves a functional purpose of inferring stable perceptual representations in a noisy, changing visual world (Ceylan et al., 2021; Cicchini et al., 2018, 2024; Collins, 2019; Fischer & Whitney, 2014; Fritsche et al., 2020; John-Saaltink et al., 2016; Manassi et al., 2023; Manassi & Whitney, 2024; Pascucci & Plomp, 2021; Pascucci et al., 2023; van Bergen & Jehee, 2019). One such account proposes that serial dependence effects reflect spatially tuned continuity fields in visual cortex, which affect consecutive stimuli activating neural populations which overlap in feature sensitivity (Fischer & Whitney, 2014; Manassi & Whitney, 2024; Manassi et al., 2023). In line with this account, we observed a decrease in serial dependence with increasing spatial distance between items.

Nevertheless, we also observed that serial dependence was still stronger for items at *different* peripheral locations (i.e., right peripheral followed by left peripheral and vice versa, and thus spanning a longer distance) than for central items. One way in which the continuity field account explains the stronger biases in the periphery is that the stronger pooling of signals results in extensively overlapping spatial receptive fields as well as feature sensitivity, which enables stronger integration of signals not only across space, but also across consecutive perceptual instances of similar features.

Other accounts attribute serial dependence effects to decision making processes that may play out at either perceptual or post-perceptual levels (Ceylan et al., 2021; Cicchini et al., 2018; Fritsche et al., 2020; Gallagher & Benton, 2022; Pascucci et al., 2019; van Bergen & Jehee, 2019; You et al., 2023). Most of these decision accounts put the emphasis on the uncertainty or imprecision of the perceptual input as the major contributor to serial dependence, though Pascucci et al. (2019) proposed an account in terms of continuity fields also at the decision level. The idea then is that the more uncertain or imprecise is the current input, the stronger the relative weight of the preceding evidence in shaping the current percept (and this weighting may further differ on the basis of factors such as stimulus reliability and time scale; e.g., Cicchini et al., 2018; Fritsche et al., 2020; van Bergen & Jehee, 2019; You et al., 2023). Decision accounts can then explain the current data by assuming that peripheral vision is less precise and thus relies more strongly on the previous input. Our study cannot judge between perceptual and decision accounts, but whether perceptual or decisional in nature, any account based on uncertainty would put even more weight on the preceding evidence if that evidence were more reliable and more precise. A priori one would thus predict the strongest serial dependence for peripheral stimuli following central stimuli—given the high fidelity of the latter. This is not what we observed, the strength of the serial dependence was impervious to the previous stimulus eccentricity. This could mean that decisions in our experiments were solely based on the uncertainty of the current stimulus (cf. Ceylan et al., 2021; Gallagher & Benton, 2022; You et al., 2023), or that prior feature evidence is weighted more strongly when spatially congruent with the current evidence compared with when spatially incongruent.

Alternatively, the finding that the eccentricity on the previous trial did not matter, may have been caused by the fact that the probe stimulus (which was the last stimulus on each trial and the one participants responded to) was always presented centrally, and hence the bias as caused by the preceding trial was mainly based on this central stimulus. However, one aspect of our findings makes this explanation less likely to be the whole story, as the strength of the serial dependence was affected by whether the previous stimulus was in the same location or not. Specifically, in Experiment 1, peripheral → peripheral dependence was stronger than central → peripheral dependence, and stronger than peripheral → peripheral (switch). Moreover, in Experiment 3, peripheral → peripheral (switch) was stronger than central → peripheral dependence. None of these effects would be predicted if all that mattered on the preceding trial was the central probe stimulus. While we cannot exclude the possibility that the central probe too may have acted as a source of serial dependence for the next trial, it does not as such explain why serial dependence is then stronger for peripheral stimuli on that trial. As discussed, serial dependence has been demonstrated to be strongest for spatially overlapping stimuli, and so the probe should have led to stronger effects for central stimuli on the subsequent trial. If anything, this may thus have led to an underestimation of the increased peripheral dependence effects.

We can exclude one potential other account though, in terms of remapping orientation information from periphery to fovea. It has been argued that visual stability is also desired *between* peripheral and foveal vision, specifically, across eye movements (Golomb & Mazer, 2021; Melcher, 2007; Rolfs et al., 2011; Stewart et al., 2020). Specifically, peripheral vision delivers advance information on what that target will look like once foveated (Herwig & Schneider, 2014), thus supporting perceptual continuity. Relatedly, several studies suggest that peripheral items can be represented by regions normally involved in processing foveal stimuli (Oletto et al., 2023; Stewart et al., 2020; Weldon et al., 2016; Williams et al., 2008). Furthermore, the fact that the probe was always presented centrally may have promoted such remapping of information. While we do not argue against these accounts, our data suggests that serial dependence does not serve a crucial role in this process, as it would predict strongest biases for central stimuli following peripheral stimuli, which is not what we observed.

Our findings also contribute to the ongoing debate over whether serial dependence is anchored in retinotopic or allocentric space. When retinotopic and allocentric positions were dissociated—such as by tilting the head while presenting stimuli foveally—serial dependence aligned with allocentric coordinates (Mikellidou et al., 2021), supporting the idea that bias arises from tuning of bottom-up sensory neurons to temporally and spatially proximal inputs. However, when similar spatial manipulations were extended into the periphery, such as through saccadic shifts, serial dependence appeared to follow retinotopic coordinates (Collins, 2019). While both reference frames likely contribute to serial dependence (Fischer & Whitney, 2014), our results show an asymmetry, as switching from fovea to periphery elicited weaker attraction than switching from periphery to fovea, despite equal allocentric distances between stimuli. This asymmetry suggests that one must be careful when investigating effects of distance or spatial shifting if these also involve eccentricity differences, as the eccentricity-related changes in serial dependence may enhance the contribution of retinotopic influences.

Note that we did not correct for perceptual discriminability differences associated with central and peripheral stimuli, for several reasons. First and foremost, the inhomogeneity of the visual field was the exact reason for conducting this study. That is, the fact that peripheral stimuli are less distinguishable (as evidenced by larger overall report errors) provides the rationale for why serial dependence may be stronger, and it was our goal to test this hypothesis. Correcting for discriminability would undermine that goal. Second, while researchers in labs may try to correct for discriminability, our everyday visual world typically does not. If we wish to know how serial biases may affect our perception in natural vision, we should not correct for its inhomogeneity.[1] However, this does leave open an alternative account in which it is not the inhomogeneity which is the crucial factor in the differential serial dependence, but eccentricity per se (i.e., purely the spatial distance from fixation, regardless of the changes in discriminability that comes with it). We currently cannot test for this possibility, as this would require equalizing central and peripheral vision. Note that that is far from easy to do. While increasing stimulus size through so-called *M-scaling* has been suggested (Azzopardi & Cowey, 1993; Cowey & Rolls, 1974; Daniel & Whitteridge, 1961; Horton & Hoyt, 1991; Rovamo & Virsu, 1979), even simple stimuli like oriented gratings come with multiple properties, namely orientation, spatial frequency, and contrast—each of which may follow a different discriminability profile across eccentricity, and which may moreover combine in complex convoluted ways that may again differ across eccentricity (Strasburger et al., 2011), and likely differs per individual. The same would go for adding different noise levels (where the noise properties too would come with certain eccentricity profiles). These different perceptual dimensions and their interactions would each have to somehow be corrected 'equally', otherwise any observed differences in serial dependence (or lack thereof) could be attributed to the stimulus change rather than changes in the visual system (which is what we are interested in). Given that ultimately these properties all map onto the same performance measure, this is a real challenge. While successfully doing so would indeed help answer the question whether it is eccentricity per se or the discriminability that comes with it that is important, we consider this endeavour beyond the scope of the current study. What we can conclude from our study is that under normal (i.e., uncorrected) visual circumstances, more peripheral stimuli are subject to stronger serial dependence.

Experiments 2 and 3 indicate that attention was not a major contributor to the main findings. In Experiment 2 we provided directional cues with 100% valid information, and compared these to noninformative cues. In Experiment 3 we balanced the proportion of central and peripheral stimuli. Neither of these manipulations changed the finding that peripheral stimuli come with stronger biases. It may come as a surprise that we found little evidence for a top-down (attentional) difference between central and peripheral processing in the current experiment, especially in the light of earlier work pointing towards eccentricity-dependent attention effects (Intriligator & Cavanagh, 2001; van Heusden et al., 2021, 2023, 2024; Wolfe et al., 1998; Yeshurun & Carrasco, 1998), and popular theoretical frameworks suggesting that attention may be the means by which VWM retains its information (e.g., Adam et al., 2022; Kiyonaga & Egner, 2013; Olivers, 2008). However, several studies now suggest that during VWM maintenance peripheral information suffers relatively little additional deterioration beyond the sensory input itself (Harrison & Bays, 2018; Kandemir & Olivers, 2024). For example, Kandemir and Olivers (2024) observed equally strong decoding signals during maintenance of the central and peripheral stimuli also used here. If top-down resources get more or less evenly distributed during VWM maintenance, then this explains why trying to manipulate these resources had little effect on serial dependence in our experiments.

Furthermore, we found in Experiment 2 that if anything it was the *central* item that benefited from cueing. This provides some support for the idea that observers were already attentionally biased towards the peripheral items, which would then result in weaker cueing effects. In Experiment 3 we corrected for this and similar results were still observed. We conclude that in our study, the enhanced serial biases observed for peripheral vision were not sensitive to attention. Note that this is inconsistent with an uncertainty account as our attention manipulations did improve overall discriminability (and thus presumably reduce uncertainty). This is not to say that attention plays no role in serial dependence, as there is evidence that it does (Fischer & Whitney, 2014; Manassi et al., 2023), but it seems rather modulatory and not essential, and serial dependence appears largely an automatic effect (Cicchini et al., 2024). Future studies may employ more powerful attention manipulations to assess if these then do interact with eccentricity in serial dependence.

In sum, peripheral vision not only differs from central vision in terms of the quality of the feedforward sensory processes and the efficacy of top-down feedback mechanisms, but, as we show here, also in terms of the effects of perceptual history—where peripheral vision relies more on the past than central vision.

[1] This is not to claim that our stimuli and task were naturalistic.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.3758/s13414-025-03208-1.

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Author contributions C.N.L.O. and G.K. conceptualized the study. G.K. programmed the experiments, managed data collection, processed and analysed the data. C.N.L.O. and G.K. wrote the paper.

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Data availability The data and materials for all experiments are available **Code availability.** The scripts for the experiments and analyses are available at: https://osf.io/v56hn/?view_only=6d4d5bba493b4bc788c3eed8decd8370

Declarations

Competing interests The authors have no competing or conflicting interests to declare.

Ethics approval The studies were conducted under umbrella protocol VCWE-2021–009 as approved by the Scientific and Ethics Review Board of the Faculty of Behavioural and Movement Sciences at the Vrije Universiteit Consent to participate. Participants provided informed consent prior to participation.

Consent for publication Participants provided consent for the public sharing and use of the anonymized research data. No personal data are being published or publicly shared at https://osf.io/v56hn/?view_only=6d4d5bba493b4bc788c3eed8decd8370.

Open practices statement The data, scripts for analyses and the output of analyses for all experiments are available online (https://osf.io/v56hn/?view_only=6d4d5bba493b4bc788c3eed8decd8370). Neither of the experiments were preregistered.

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